FDS EXPERIMENT-17

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Experiment to understand Linear Regression for a given data set.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
```

```
data = {
    "Study_Hours" : np.array([1,2,3,4,5]),
    "Marks" : [35,40,50,55,65]
}

df = pd.DataFrame(data)
print(df)
```

```
   Study_Hours  Marks
0            1     35
1            2     40
2            3     50
3            4     55
4            5     65
```

```
X = df[["Study_Hours"]]
Y = df["Marks"]
```

```
model = LinearRegression()
model.fit(X,Y)

print("Slope:",model.coef_[0])
print("Intercept:",model.intercept_)
```

```
Slope: 7.500000000000001
Intercept: 26.499999999999996
```

```
predicted_marks = model.predict([[8]])
print("Predicted marks:",predicted_marks[0])
```

```
Predicted marks: 86.5
```